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Hidden results of functional diversity in macroinvertebrates: response of trait groups to flow intermittency in lowland streams

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ABSTRACT

As flow intermittency increases globally, understanding how macroinvertebrates respond at a functional level in streams becomes crucial for effective water management. This study investigates the functional response of macroinvertebrates to flow intermittency in Hungarian lowland streams over a 10-month period, comparing intermittent and perennial streams across the entire community, trait groups, and trait states. We aimed to demonstrate differences in functional diversity using metrics such as functional richness, evenness, divergence, dispersion, and Rao's quadratic entropy. For trait groups, we focused on both response and stable traits, anticipating greater differences in response traits. Additionally, we identified which trait states, as indicated by community-weighted means, contribute significantly to variations between streams. Our findings indicate that functional diversity in intermittent streams is generally lower or equivalent to that in perennial streams, suggesting disparities in functional trait composition. Significant differences were observed within trait groups, indicating specific responses of trait groups to intermittency. We identified resilience trait states—including current velocity, reproduction, respiration, female wing length, locomotion and substrate relation, aquatic stages, and resistance forms—that contributed to these differences. The 3-level analysis revealed a hidden relationship: differences at the state level do not consistently manifest at the trait group or whole community levels. Investigating functional changes at the trait group or trait state level can serve as an early indicator of community shifts. Incorporating trait groups as indices into assessment frameworks could facilitate early detection of functional changes during bioassessment. We recommend considering trait groups when evaluating functional responses to flow intermittency in future studies.

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Introduction

Most watercourses worldwide are already intermittent, and this intermittency is increasing because of the impacts of climate change and human activities (Messager et al. 2021). In numerous instances, perennial streams are experiencing flow intermittency attributed to rising temperatures, uneven precipitation (Pendergrass and Knutti 2018), and escalating water demand (Dyer et al. 2013). Currently, concern is growing about the large, increasing number of intermittent streams (Messager et al. 2021) worldwide and the continental climate. Instability in water condition poses significant risks to aquatic biodiversity, disrupting

ecological processes and changing the underlying mechanisms of community assembly (Cazzolla Gatti 2016).

The Water Framework Directive (WFD) is a European Union legislation aimed to enhance and safeguard the quality of surface waters and their ecological status. In accordance with the requirements of the WFD, biological indices have been developed for ecological assessment based on accumulated ecological knowledge about the pressures on the communities as well as their structural and functional responses (Hering et al. 2006, Birk et al. 2012). For many years, these complex and intercalibrated indices have been used in most countries

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for bioassessment in River Basin Management Plans (Boda et al. 2023). Because intermittent or temporary streams are absent in the European river type classification in the temperate zone, these widely used indices were developed mostly for perennial waterbodies; therefore, the newly occurring flow intermittency poses new challenges to water policy (Solheim et al. 2019, Trambly et al. 2020). The initial and most crucial step involves the establishment of a comprehensive typology (Datry et al. 2014) that includes the classification of waterbodies affected by drying based on their unique water conditions or the intermittent nature of their water regimes. Subsequently, conducting a thorough biological validation of these diverse water types using macroinvertebrates is imperative (Miliša et al. 2022). The biological validation requires numerous field experiences to determine how taxa, taxa groups, traits, and trait groups react to drying and which structural or functional metrics and indicators are the most effective to measure the structural and functional response to flow intermittency.

Flow intermittency plays a crucial role in shaping the species composition of aquatic communities, affecting both local species richness and species turnover across landscapes (Sabater et al. 2016). The taxonomic diversity of macroinvertebrates tends to be lower in intermittent streams compared to perennial streams, regardless of regional, seasonal, spatial and stream type differences (Soria et al. 2017, Carey et al. 2023). As a result of time-sharing communities, intermittent waters exhibit greater seasonal variations in taxonomic composition, abundance, and biological characteristics (Bêche et al. 2006, Bogan and Lytle 2007). Additionally, beta diversity often increases in intermittent systems as species compositions vary more drastically across habitats, reflecting the diverse niches created by fluctuating flow regimes (Sánchez-Montoya et al. 2018).

Because of the geographic differences in regional species pools and taxa richness among the drought-related studies, the most general conclusions regarding the effect of flow intermittency on macroinvertebrate community can be drawn from the results of the functional aspects (Mouchet et al. 2010, Schmera et al. 2023). Flow intermittency is known to influence the functional diversity of aquatic communities by exerting selective pressure on traits related to resistance and resilience (Bozóki et al. 2024). Most previous studies on the functional diversity of macroinvertebrates provided contrasting results between intermittent and perennial streams based on biogeographical context, spatial scale, and variation in flow intermittency (Bêche et al. 2006, Bogan and Lytle 2007). Therefore, when considering a whole year and all of its seasons, functional diversity may be higher in intermittent waters than perennial

waters because of the distinct functional traits of taxa associated with different hydrological phases (Bonada et al. 2007). The higher functional diversity can be derived from the detectable differences of seasonality and cycling hydrological phases (Bêche et al. 2006), although other studies highlighted lower functional richness and functional evenness in intermittent waters because of its desiccation-sensitive traits (Ledger et al. 2013, Schriever et al. 2015). Especially, differences in functional diversity, richness, and redundancy among different hydrological water regimes have not been observed at small spatial scales and across various climates (Leigh et al. 2016).

Different strategies promote and determine the ability and potential to survive a flow intermittency. Taxa with traits that confer resistance to drying events, such as desiccation tolerance or the ability to enter diapause, are favored in intermittent streams, contributing to the community's ability to persist during dry events (Crabot et al. 2021). Conversely, traits associated with resilience, such as rapid growth rates or high dispersal capabilities, become advantageous for recolonization and population recovery once water returns (Sarremejane et al. 2021). In newly drying stream sections, the traits associated with resistance, not resilience, structure the macroinvertebrate community (Bozóki et al. 2024). However, life history traits alone seem to be poor predictors of species response to flow intermittency (Carey et al. 2021). Less is known about how ecological, dispersal, and biological trait groups, including their respective traits and trait states, react to drying (Strachan et al. 2015). There is a growing need to understand the specific traits and trait groups that can shape the macroinvertebrate communities of intermittent streams from a functional perspective (Kelso and Entekin 2018).

While numerous studies have explored the functional diversity of macroinvertebrates in intermittent mid to high altitude streams (>200 m a.s.l.; Bêche et al. 2006, Bogan and Lytle 2007, Schriever et al. 2015, Leigh et al. 2016), research focusing on lowland intermittent streams remains notably scarce (Verdonshot et al. 2015). These variable topographical profiles of stream habitats can give different and even apparently contradictory results in the composition of benthic diatom assemblages (Falasco et al. 2016, B-Béres et al. 2019). Lowland streams differ from headwaters in several ways: they have much lower flow velocity, generally deeper water, mostly straightened channels, lower habitat diversity, greater vegetation cover, and fine particulate streambeds (Borics et al. 2016). These environmental conditions collectively lead to slower material transport and increased organic

matter input (Fisher and Likens 1973, Vannote et al. 1980). Additionally, lowland streams are characterized by higher connectivity because of the lack of geographical barriers, less forest cover, and significant evapotranspiration (Leibowitz et al. 2018). These differences significantly influence the biotic factors that determine the structure of biotic communities in lowland streams compared to headwaters (Allan et al. 2007, Dodds and Oakes 2008). In these lowland streams, anthropogenic impacts are much more pronounced because these streams provide ecosystem services in densely populated areas, and significant water abstraction is used to irrigate the surrounding agricultural landscape. Although drying may be a more frequent phenomenon in the upper, hilly parts of a catchment (Lake 2003), aquatic communities in lowland regions are also affected by the effects of flow intermittency. Revealing the detailed functional response of macroinvertebrate communities to flow intermittency in lowland streams, highly important from a human perspective, can help develop strategies that support water policy and the biological validation of intermittent stream types.

The use of functional traits in metrics provides several perspectives for measuring the functional diversity of communities, including the use of different trait groups (which integrate traits) to indicate possible responses to disturbance based on their unique characteristics. This study aimed to assess how various metrics and trait groups can help detect the response of macroinvertebrate assemblages to flow intermittency in lowland streams. We characterized communities to evaluate their response at the trait group, trait, and trait state (the different modalities of traits) levels. To achieve this, we surveyed macroinvertebrate assemblages monthly from spring to winter in Hungarian lowland streams.

We hypothesized that, regarding the whole community and measured by any functional metric, functional diversity in intermittent lowland streams would be lower or equivalent compared to perennial ones. At the trait group level, categorized into response (Ecological) and stable (Biological, Dispersal, and Life-history) traits, we hypothesized different responses between intermittent and perennial streams. We assumed that the response trait group would show greater differences between intermittent and perennial streams across all metrics compared to the stable trait group. Additionally, we aimed to determine which trait state contributes most to the observed differences at both the trait and trait group levels between intermittent and perennial streams.

Materials and methods

Study area

The investigated streams are located in Hungary within the Great Hungarian Plain, which encompasses the largest geographic area in the Carpathian Basin. During the last century, mean annual temperature increased by 1 °C in the continental region of the Carpathian Basin (Bartholy et al. 2014) and 1.7 °C in Hungary while annual precipitation decreased by 10%, with altered precipitation patterns (IEA 2021). These changes led to altered regimes in streams, resulting in an increased number of temporary streams in the area (B-Béres et al. 2019). Approximately 50% of the Hungarian waterbodies monitored under the WFD are located in this area. Agricultural land use is the main economic activity and its associated impacts, for example water abstraction, which combined with lower forest cover increases evapotranspiration rates. Small lowland streams are typically straight with low slopes, belonging to the R-05 broad river type category (Solheim et al. 2019). Therefore, the hydrological regimes of these streams frequently alternate between the flowing, non-flowing and dry phases. Streambed exhibits a high abundance of macrophytes, and the substrate comprises mostly fine particulate organic matter (FPOM) and coarse particulate organic matter (CPOM) derived from plant debris, especially in areas with low water levels (Borics et al. 2016, Kiss et al. 2024). The streams shared identical environmental conditions; no differences were observed among them in terms of measured background variables such as flow velocity, water depth, channel width, vegetation cover, and streambed habitats. Based on the Hungarian classification of running waters using the criteria established by the EU WFD, streams in our survey belong to the same waterbody type with the same environmental conditions. Additionally, because of the small geographical distance among the sampled streams, they share a regional species pool (Boda et al. 2023). We investigated both intermittent and perennial streams. Intermittent streams exhibited alternating flow patterns, including phases of flowing, nonflowing, and dry conditions, with variations in the frequency and duration of drying events. By contrast, perennial streams were characterized by stable hydrological regimes and a continuous flowing phase.

Sampling design

We studied 8 streams, 3 perennial and 5 intermittent (Fig. 1). Sampling occurred monthly from March to December 2020. Macroinvertebrate samples were collected during

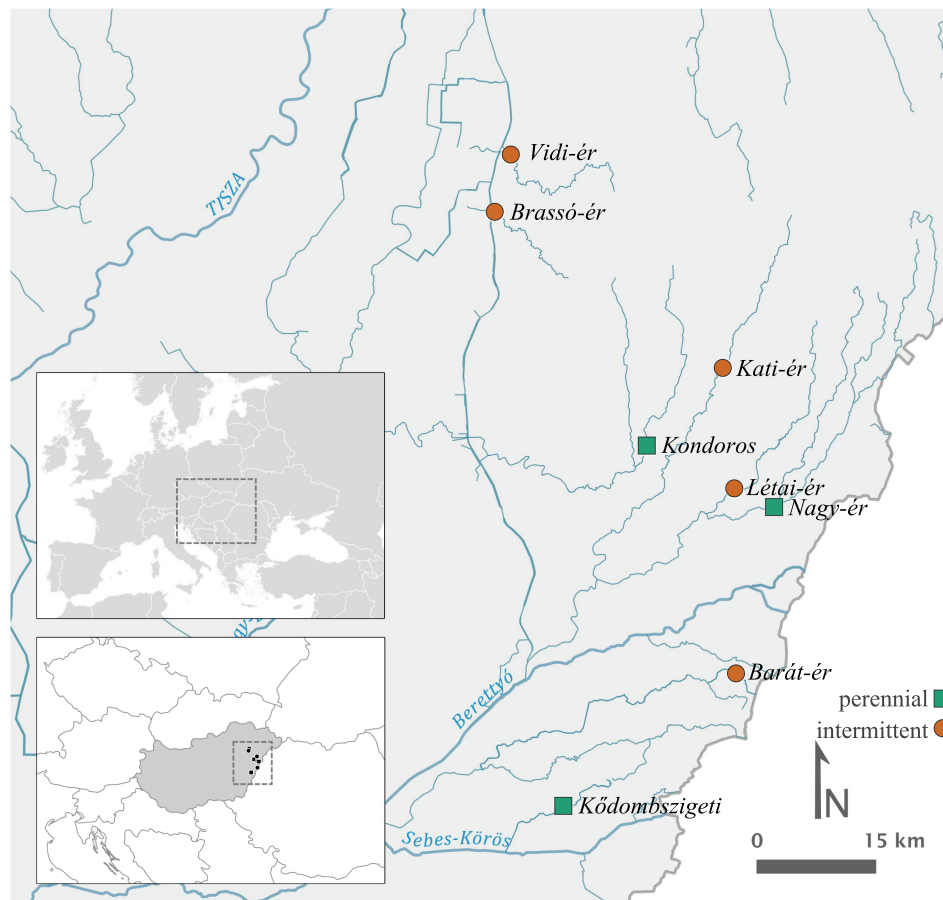


Figure 1. Schematic map of the sampling area (Hungary, Europe) showing perennial and intermittent streams.

any wet phase, defined by the presence of water, regardless of whether the water was actively flowing or stagnant; sampling was not conducted when the streambed was dry. We collected 55 samples during the sampling campaign, comprising 30 samples from perennial streams and 25 samples from intermittent streams. In addition to the dates of biological samplings, we recorded the wet or dry phases in the streambed. For intermittent streams, dry phases were recorded on 1–8 occasions, but no stream remained completely dry throughout the sampling campaign (Supplemental Table S1). For perennial streams, wet phases were recorded on all sampling occasions.

We utilized a multihabitat sampling approach combined with a standardized “kick and sweep” technique, employing a hand net with a 1 mm mesh size and 25 cm × 25 cm frame, following the Hungarian sampling protocol (Boda et al. 2023). Five sample units covering an area of 0.3125 m² were collected from each sampling site to form the composite sample. Samples were taken from all habitat types, including emergent and submergent macrophytes, CPOM, FPOM, macroalgae, and stones. For the macroinvertebrate analysis, abundances were standardized to individuals per square

meter. Samples were preserved in 70% ethanol until sorting and identification. Our study assessed 11 key macroinvertebrate groups: Gastropoda, Bivalvia, Hirudinea, Crustacea, Ephemeroptera, Plecoptera, Odonata, Trichoptera, Heteroptera, Coleoptera, and Diptera. Identification to the lowest possible taxonomic level was accomplished using the keys provided by Lillehammer (1988), Savage (1989), Haybach (1999), Csabai (2000), Bauernfeind et al. (2001), Eggers and Martens (2001), Andrikovics and Murányi (2002), Csabai et al. (2002), Kontschán et al. (2002), Krno (2004), Zwick (2004), Eiseler (2005), Waringer and Graf (2011), Bauernfeind and Soldán (2012), Dobson (2012), Amburus (2018).

Traits of macroinvertebrates

Two databases, DISPERSE (Sarremejane et al. 2020) and Freshwater Ecology (FWE; Schmidt-Kloiber and Hering 2015), were used to construct our drought-related trait database. To eliminate differences between the 2 databases and to achieve the highest trait coverage, we reduced our taxonomic data to the genus level. To mitigate family-level data gaps (e.g., Chironomidae), trait

values were derived from the average value of the genera within the family. For certain species, however, such as *Metreletus balcanicus* (Ulmer, 1920), trait data were absent at the genus level; therefore, we used the trait values at a species level within the study area.

We standardized trait states of databases in our trait matrix (genus $\times$ trait state) to a value between 0 and 1 for each genus using fuzzy coding (Chevene et al. 1994, Kefford et al. 2020) and used the sum of trait values for standardization (Schmera et al. 2014), with missing data set to a value of 0. Our categorization was based on the Biological and Ecological trait groups from the FWE database. The Biological traits of the FWE database were further classified into more detailed trait groups, considering DISPERSE and according to the work of Usseglio-Polatera et al. (2000). Thus, the traits were categorized into 4 trait groups: Ecological, Biological, Dispersal, and Life-history (Table 1).

Data analysis

The statistical analyses involved calculating the biotic and the trait matrices. The biotic matrix was compiled by creating a data table for each sampling occasion $\times$ genus, containing individuals per square meter values. Standardization was performed within each trait group, considering the number of traits before calculating functional diversity components and community-weighted mean (CWM) values for the trait matrix (genus $\times$ trait state).

To compare the community, we used the *FD* package in R to calculate CWM values of selected genus traits (Laliberté et al. 2014) to represent the mean of genus trait values at each sampling occasion, taking into account the relative taxa abundance (Ricotta and

Moretti 2011). Analysis of similarities (ANOSIM) was performed to assess differences in the CWM values distribution in the 2 water types for each trait group. The separation of the 2 water types based on CWM values was also illustrated with non-metric multidimensional scaling (NMDS) using the *vegan* package (Oksanen et al. 2020) in R software 4.0.5. (R Development Core Team 2021).

To compare the functional diversity of intermittent and perennial streams, functional richness (FRic), functional evenness (FEve), functional divergence (FDiv), functional dispersion (FDis), and Rao's quadratic entropy (RaoQ) were calculated using the *dbFD* function in the *FD* package (Laliberté and Legendre 2010, Laliberté et al. 2014) in R 4.0.5. (R Development Core Team 2021). FRic was the size of the niche space occupied by the functional community, without considering the relative abundances of traits; FEve was the evenness of the distribution of trait abundances within the niche space; FDiv, the abundance-weighted values of traits at the periphery of the niche space, indicated the extent of differences between the most common and average trait values; and FDis was the spread of traits from the centroid within the niche space, weighted by the relative abundances of traits. RaoQ was the measure of pairwise differences between taxa within the functional community, incorporating the relative abundances of taxa (Botta-Dukát 2005, Mason et al. 2005, Villéger et al. 2008, Laliberté and Legendre 2010). To visualize the difference in values of functional diversity components in perennial and intermittent streams, we used boxplots with jitter utilizing the *ggstatsplot* package (Patil 2021) in R 4.0.5 (R Development Core Team 2021). The Welch test was performed to compare the means between the 2

Table 1. List of trait groups, individual traits, and their corresponding codes used in our study, along with the sources databases.

Trait group		Trait	Code of traits	Database
Ecological (Response)	Ecological	Current preference	cup	Freshwater Ecology
		Current velocity (preferendum) according to Tachet et al. (2010)	cuv	Freshwater Ecology
		Stream zonation preference	szp	Freshwater Ecology
Biological (Stable)	Biological	Feeding type	fet	Freshwater Ecology
		Reproduction according to Tachet et al. (2010)	rep	Freshwater Ecology
		Respiration according to Tachet et al. (2010)	res	Freshwater Ecology
	Dispersal	Locomotion and substrate relation according to Tachet et al. (2010)	lsr	Freshwater Ecology
		Dispersal strategy	dis	Disperse
		Female wing length (mm)	fwl	Disperse
		Maximum body size (cm)	s	Disperse
		Propensity to drift	drift	Disperse
		Wing pair type	wnb	Disperse
		Aquatic stages according to Tachet et al. (2010)	aqs	Freshwater Ecology
	Life-history	resistance forms according to Tachet et al. (2010)	ref	Freshwater Ecology
		Adult life span	life	Disperse
		Life cycle duration	cd	Disperse
		Lifelong fecundity (number of eggs per female)	egg	Disperse
		Potential number of reproductive cycles per year	cy	Disperse

groups when the values were normally distributed, which occurred for FRic, FEve, FDiv, and FDis, while the Kruskal-Wallis test to assess differences between the medians was used for RaoQ when the data were not normally distributed.

To establish the indicative traits or trait states, we utilized a generalized linear model to calculate the effect of type (perennial or intermittent) and trait states on CWM values. The results include the water-type effect, presented through univariate tests with sigma-restricted parameterization effective hypothesis decomposition, allowing the examination of the dependent variables while taking into account the differences in variance.

Additionally, the effect of trait states were confirmed using the Duncan test for significant differences within the Statistica 8.0 program (Statsoft 2008).

Results

Response at community level

The macroinvertebrate community showed significant differences across all functional levels. In the functional community level, FRich and FDiv were significantly lower in intermittent streams ($p < 0.05$; Fig. 2). Based on ANOSIM, significant differences were observed in

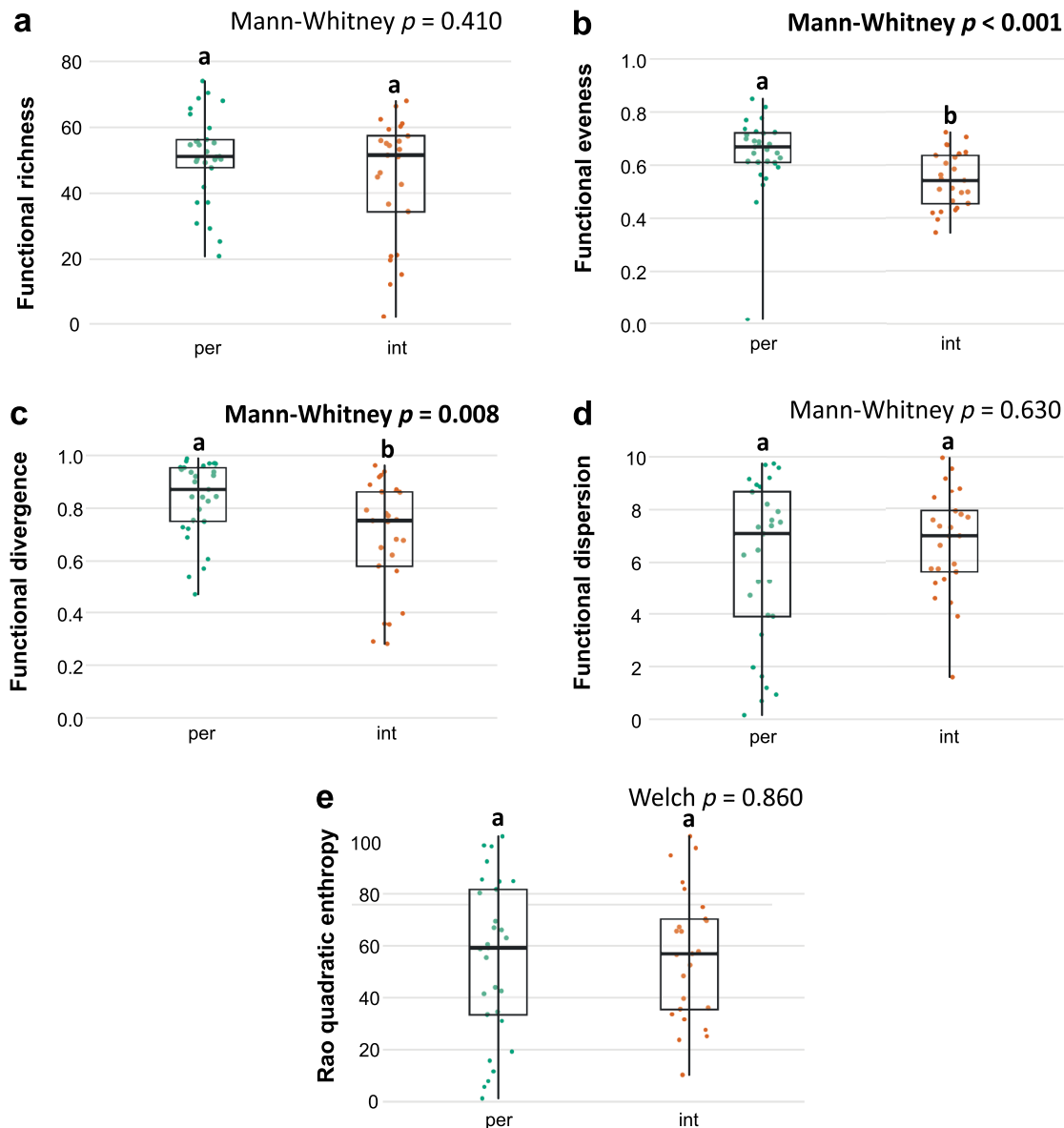


Figure 2. Differences in functional diversity metrics of communities in intermittent and perennial streams: (a) functional richness, (b) functional evenness, (c) functional divergence, (d) functional dispersion, (e) Rao's quadratic entropy. The significant differences are marked in bold at the top right corner of the plots. The central horizontal line of the box plots represents the median while RaoQ quadratic entropy indicates the mean.

Table 2. The results of the ANOSIM test, R and p values for all trait and trait groups between intermittent and perennial streams. Significant differences are marked in bold. (Bray-Curtis similarity, permutation: 9999).

	Functional community	Trait group			
		Biological	Ecological	Dispersal	Life-history
R value	0.3686	0.2993	0.5269	0.1161	0.3834
p value	0.0001	0.0001	0.0001	0.0056	0.0001

the CWM matrix between the 2 water types ($p < 0.05$ for all trait groups; Table 2).

Response at trait group level

The similarity between the 2 types of water was the highest in the case of the Dispersal trait group ($R = 0.116$), while the greatest differences were found for the Ecological trait group ($R = 0.527$), as indicated by the CWM values of trait states. The R -values of the Biological ($R = 0.299$) and Life-history ($R = 0.383$) trait groups fell between those observed in the most similar and most different trait groups (Table 2). These separations are presented in NMDS plots (stress values of trait groups were Ecological: 0.1118; Biological: 0.1503; Dispersal: 0.1135; and Life-history: 0.0876; Fig. 3).

Significant differences were observed across all trait groups between the water types, indicating distinct functional characteristics. However, variations were noted in the specific metrics analyzed. In the perennial type, FRic was significantly higher in the Ecological ($p = 0.007$), Dispersal ($p = 0.020$), and Life-history ($p = 0.002$) trait groups, and standard deviations had a wider range in the Dispersal and Life-history trait groups for intermittent streams. FEve was significantly higher in the Biological ($p = 0.050$), Ecological ($p = 0.006$), and Dispersal ($p = 0.001$) trait groups but not in the Life-history trait group (Table 3).

The comparison of perennial and intermittent water types reveals significant differences at the trait group level, particularly in Life-history, with notable variations observed across other trait groups as well. FDiv was significantly higher ($p = 0.003$) in the perennial type only

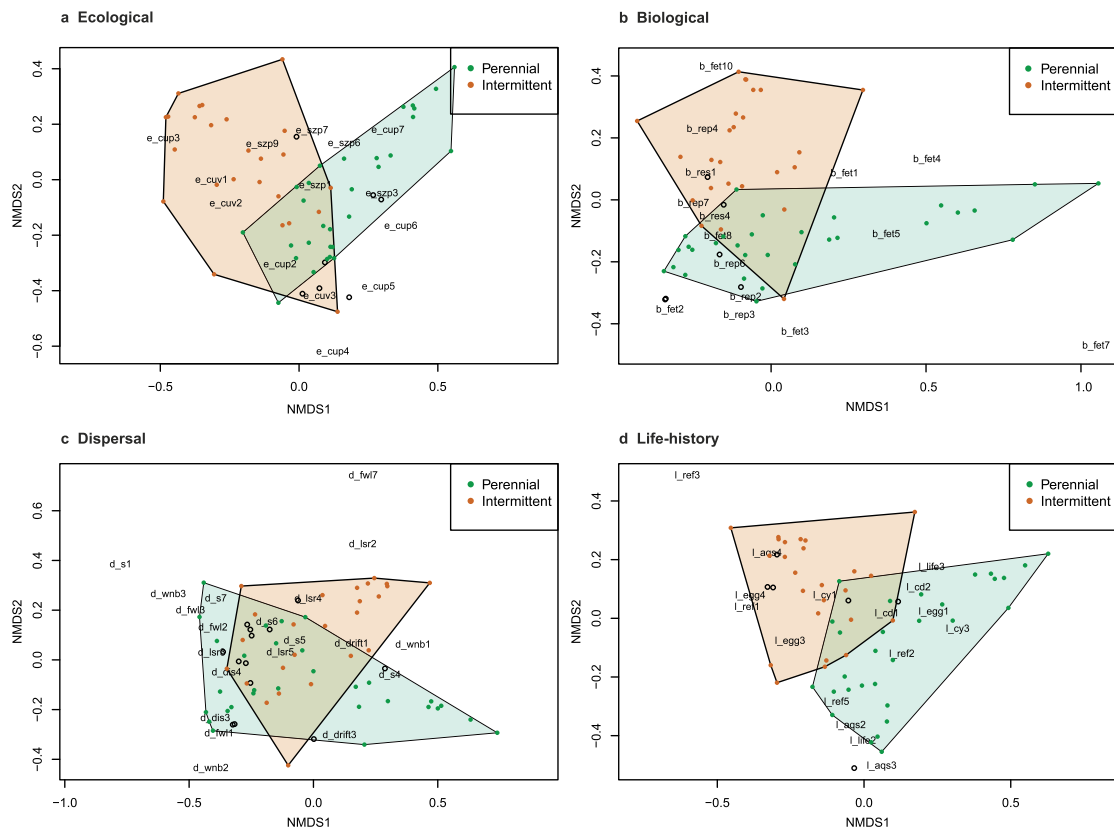


Figure 3. Non-metric multidimensional scaling (NMDS) plots for trait groups in intermittent and perennial streams. (a) Ecological, (b) Biological, (c) Dispersal, and (d) Life-history. The trait states in the trait group are shown with their trait codes in Table 1.

Table 3. Functional diversity metrics of intermittent and perennial streams trait groups: (a) functional richness, (b) functional evenness, (c) functional divergence, (d) functional dispersion, (e) Rao's quadratic entropy. For the trait groups, median, standard deviation, maximum, and minimum values of each functional diversity metrics are listed for both intermittent and perennial streams; significant differences are in bold.

	Trait groups	Type effect <i>p</i> -value	Intermittent median spread (min-max) (<i>n</i> = 25)	Perennial median spread (min-max) (<i>n</i> = 29)
a Functional richness	Ecological	0.007	16.7 ± 6.3 (3.7 - 28)	19.4 ± 8.4 (11.6 - 44.7)
	Biological	1.000	16.8 ± 7.5 (0.2 - 28.4)	17.2 ± 5.4 (5.5 - 25.8)
	Dispersal	0.020	28.8 ± 9.5 (5.5 - 39)	31.5 ± 6.9 (19.1 - 46.2)
	Life-history	0.002	16.4 ± 8.6 (1.8 - 31.6)	24.1 ± 5.2 (14.2 - 33.6)
b Functional evenness	Ecological	0.006	0.5 ± 0.2 (0.3 - 1)	0.6 ± 0.1 (0.3 - 0.8)
	Biological	0.050	0.5 ± 0.1 (0.3 - 0.7)	0.6 ± 0.2 (0 - 0.8)
	Dispersal	0.001	0.5 ± 0.1 (0.3 - 0.9)	0.6 ± 0.1 (0.2 - 0.9)
	Life-history	0.300	0.6 ± 0.1 (0.4 - 0.9)	0.6 ± 0.1 (0.1 - 0.8)
c Functional divergence	Ecological	0.630	0.8 ± 0.2 (0.3 - 1)	0.8 ± 0.2 (0.3 - 1)
	Biological	0.140	0.8 ± 0.1 (0.5 - 1)	0.7 ± 0.2 (0.1 - 0.9)
	Dispersal	0.130	0.8 ± 0.2 (0.4 - 0.9)	0.8 ± 0.2 (0.5 - 1)
	Life-history	0.030	0.7 ± 0.2 (0.4 - 0.9)	0.8 ± 0.2 (0.4 - 1)
d Functional dispersion	Ecological	0.650	3.5 ± 1.2 (0.7 - 5.5)	3.5 ± 1.6 (0.1 - 5.2)
	Biological	0.990	2.9 ± 1 (0.5 - 4.5)	3.4 ± 1.4 (0.1 - 4.7)
	Dispersal	0.900	3.9 ± 1.3 (0.6 - 5.2)	3.5 ± 1.8 (0.1 - 6.3)
	Life-history	0.070	3.6 ± 0.9 (1.2 - 5.2)	3.2 ± 1.5 (0.1 - 4.6)
e Rao quadratic entropy	Ecological	0.820	15.2 ± 8.6 (2.1 - 34)	14.2 ± 8.8 (0.4 - 29.8)
	Biological	0.630	10.8 ± 5.5 (1.6 - 21.2)	13.8 ± 7.3 (0.2 - 25.4)
	Dispersal	0.600	17.5 ± 8 (2 - 28.4)	16.9 ± 10.6 (0.4 - 41.2)
	Life-history	0.040	15.3 ± 5.7 (4.8 - 27.7)	13.2 ± 7 (0.3 - 22.2)

in the Life-history trait group. In the intermittent type, FDis was marginally ($p = 0.070$) higher while RaoQ was significantly ($p = 0.040$) higher in the Life-history trait group. Considering trait groups, the Life-history trait group exhibited significant differences in 4 out of 5 diversity metrics between the 2 water types. Meanwhile,

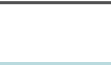
both the Dispersal and Ecological trait groups showed differences in 2 metrics each, whereas only 1 metric in the Biological trait group differed significantly between the 2 water types (Table 3).

Significant differences were detected in 7 out of the 18 traits between the 2 types of streams. Significant

distinctions were exhibited by all traits in the Biological trait group (reproduction; $p = 0.004$, respiration; $p = 0.001$) and Ecological trait group (current velocity; $p < 0.001$), depending on the type (intermittent-perennial) and trait characteristics. For the Dispersal trait group,

the locomotion and substrate relation ($p < 0.001$) and female wing length (mm; $p = 0.050$) were significantly different. Within the Life-history trait group, differences were observed in the aquatic stages ($p = 0.001$) and resistance forms ($p = 0.003$; Table 4).

Table 4. Summary of differences in community-weighted mean (CWM) values between intermittent and perennial streams. Only traits with a significant effect of the type (intermittent or perennial) are included for each trait group, along with corresponding codes (Table 1) for all trait states. Trait p values represent the effect of type in the generalized linear model (GLM) model on trait CWM. Trait state significant differences in CWM values between intermittent and perennial streams were identified by the Duncan test, highlighted in bold. Orange columns indicate higher CWM values in intermittent streams while green columns indicate higher CWM values in perennial streams. Darker bars in colored cells represent significant differences.

Trait group	Trait	type effect p -value	Trait states code	TS effect p -value	Intermittent	Perennial
Ecological	Current velocity	0.000	cuv01	0.00		
			cuv02	0.01		
			cuv03	0.42		
			cuv04	0.17		
Biological	Reproduction	0.004	rep01	0.07		
			rep02	0.72		
			rep03	0.50		
			rep04	0.00		
			rep05	0.01		
			rep06	0.71		
			rep07	0.85		
	Respiration	0.001	res01	0.00		
			res02	0.11		
			res03	0.96		
			res04	0.74		
Dispersal	Female wing length	0.050	fwl01	0.00		
			fwl02	0.04		
			fwl03	0.64		
			fwl04	0.77		
			fwl05	0.99		
			fwl06	1.00		
			fwl07	0.99		
	Locomotion and substrate relation	0.000	lsr01	0.92		
			lsr02	0.00		
			lsr03	0.00		
			lsr04	0.00		
			lsr05	0.67		
			lsr06	0.58		
			lsr07	0.43		
Life-history	Aquatic stages	0.001	aqs01	0.00		
			aqs02	0.01		
			aqs03	0.09		
			aqs04	0.00		
	Resistance forms	0.003	ref01	0.22		
			ref02	0.60		
			ref03	0.00		
			ref04	0.00		
			ref05	0.00		

 Significantly higher CWM value in intermittent streams
 Higher CWM value in intermittent streams
 Significantly higher CWM value in perennial streams
 Higher CWM value in perennial streams

Response at trait state level

The CWM values of trait states within a specified trait also revealed noticeable differences (Table 4). We found 36 different CWM values in the stream types, of which 16 trait states were significant in the CWM values based on the Duncan test. Altogether, 10 trait states were higher in the intermittent types and 6 trait states were higher in perennial. Additionally, several trait states within traits showed significant results, but these trait states differently dominated the functional community in intermittent and permanent waterbodies. However, for some trait states (current velocity, female wing length), traits had higher CWM values in only 1 type. The highest number of different trait states (9) were found in the Life-history trait group (Table 4).

For the Life-history trait group, significant differences ($p < 0.001$) were observed in 3 trait states of the resistance forms trait: cells against desiccation (ref3) and diapause or dormancy (ref4), both of which had higher CWM values intermittent streams in contrast to none (ref5), which was found in higher proportions in the perennial type. Significant differences in the aquatic stages trait were detected in the following states: egg (aqs1) and adult (aqs4) states ($p < 0.001$), which had higher values in the intermittent type, in contrast to the larva (aqs2, $p = 0.010$) state, which was more pronounced in the perennial type (Table 4).

For the Biological trait group, a significant difference between the 2 types was observed only in 2 states of the reproduction trait states related to clutches, namely in case of states cemented or fixed (rep04, $p < 0.001$) and free (rep05, $p = 0.010$). Furthermore, the respiration trait showed a difference in the tegument (res01, $p < 0.001$) state (Table 4). In the Ecological trait group, only 1 trait, current velocity, exhibited a significant difference in 2 of its states: null (cuv1, $p < 0.001$) and slow (cuv2, $p = 0.010$) between intermittent and perennial water types (Table 4). The trait states of surface swimmer (lsr2, $p < 0.001$), swimmer (lsr3, $p < 0.001$), and crawler (lsr4, $p < 0.001$) in the locomotion and substrate related trait and 2 small size range states of female wing length (mm) < 5 (fwl1, $p < 0.001$) and ≥ 5 –10 (fwl2, $p = 0.04$) showed significant differences in the Dispersal trait group (Table 4).

Discussion

Our understanding of structural responses of macroinvertebrate communities to flow intermittency is extensive, but functional responses, especially in lowland

streams, are less understood. A 10-month sampling campaign aimed to identify key functional metrics, trait groups, and trait states that differentiate community responses between perennial and intermittent streams. Our hypothesis that functional diversity is lower in intermittent lowland streams compared to perennial ones was supported; we observed lower or equivalent functional diversity across all metrics in the entire community of intermittent streams, suggesting that intermittent streams may retain communities with less diverse functional traits compared to perennial streams. Furthermore, our results highlight the importance of considering different trait groups when assessing community responses. However, the differences were most pronounced in the Ecological trait group (Table 2). We also detected trait states with changes in proportions expressed in the differences observed at the trait group level. Based on these findings, we identified the functional metric components, trait groups, and trait states most suitable for subsequent index development processes.

Response at community level

Drought-related disturbances typically lead to a reduction in structural biological attributes, such as taxa richness and abundance (Arscott et al. 2010, Bogan et al. 2013, Stubbington et al. 2017, White et al. 2018). This result implies the loss of less-resistant or less-resilient taxa from macroinvertebrate communities (Chase 2007), resulting in alterations in functional diversity. Overall, functional diversity may exhibit higher levels in intermittent waters compared to perennial waters because of the distinct functional traits associated with taxa across different hydrological phases (Bonada et al. 2007) that are typical characteristics of intermittent waters. Findings of Schriever et al. (2015), however, suggest that FRich and FEve (as well as taxonomic richness) are lower in intermittent streams. At small spatial scales, no evident differences in functional diversity, richness, and redundancy have been observed across different climatic conditions (Leigh et al. 2016). Our study shows that functional diversity was marginally lower in intermittent lowland streams than perennial streams, as measured by each metric, although this difference was significant only in FEve and FDiv. The uneven distribution of functional traits in intermittent streams may indicate a lack of functional redundancy, with certain traits disproportionately represented by only a few species. In these cases, the community may be more sensitive to changing environmental conditions or disturbances because the loss of a single species with a unique functional trait

could significantly impact ecosystem processes and stability. Simultaneously, lower FDiv suggests that ecological strategies of species in intermittent streams may have converged, and as a result, the community may become more homogeneous in functional composition (Piano et al. 2020), emphasizing the ability of these metrics to capture functional differences within the community and their specificity on trait evenness and divergence. In intermittent areas, constant trait numbers result in more uniformity (Sánchez-Montoya et al. 2018). Our findings suggest that intermittency in lowland streams may reduce functional diversity, potentially leading to the dominance of standing water species (cuv1 taxa) within the assemblages. Factors contributing to this decrease could include the ability of standing water species to tolerate fluctuations in water levels and to adapt to changing flow dynamics, giving them a competitive advantage over species that are dependent on flowing water for survival (Bunn and Arthington 2002, Leigh et al. 2016).

Response at trait group level

Analyzing the trait groups respectively revealed that the specific response at trait group level drives the observed differences in the whole trait community.

In the case of FDis, no significant difference was observed between perennial and intermittent streams, either at the whole trait community or trait group level. The most plausible explanation for the similar FDis values is that the intermittent stream community does not exclusively represent a subset of the perennial stream community. While intermittent streams may share certain functional traits with perennial streams, they also exhibit additional traits or variations influenced by intermittent hydrological conditions. Consequently, differences in specific traits or trait distributions, as well as overlaps between traits, commonly resulted in almost equal overall FDis values between the 2 stream types.

No significant differences for FRich and RaoQ were observed at the whole functional community level in intermittent streams compared to perennial streams, although significant differences can be found at the certain trait group level. This phenomenon could be attributed to several factors: (1) Individual trait groups may show significant differences between intermittent and perennial streams, even if these differences may not necessarily translate into significant variations in the whole functional community (Datry et al. 2014). (2) This discrepancy could arise if the differences observed in individual trait groups are counterbalanced by

compensatory changes (Larned et al. 2010) in other trait groups, resulting in similar overall functional compositions between 2 stream types. The lack of significance at the whole functional community level may be influenced by the inherent complexity and interdependence of functional traits. Even if all trait groups exhibit significant differences between intermittent and perennial streams, the whole functional community may still maintain a relatively similar functional structure because of compensatory changes or functional redundancy within the community. (3) Because of the diverse hydrological patterns such as occurrences and duration of drying events, a characteristic of intermittent streams, the variability of the metric values is higher within these streams compared to perennial streams (Dewson et al. 2007).

By contrast, differences in individual trait groups led to differences in the whole trait community, as for the FEve and FDiv metrics. In these cases, the trait groups responsible for these differences can be detected; these traits for FEve include current velocity, locomotion and substrate relation, female wing length, reproduction, and respiration, whereas the relevant traits for FDiv are aquatic stages and resistance forms. The results indicated that FEve was significantly lower in intermittent than perennial streams, both in the whole community and individual trait groups, such as Ecological, Biological, and Dispersal. For FEve, only the Life-history trait group failed to significantly manifest the difference, whereas, for FDiv only the Life-history trait group had a significant impact on shaping functional diversity in response to flow intermittency. By contrast, our analysis revealed significant differences in FRich, FDiv, and RaoQ between perennial and intermittent streams, particularly within the Life-history trait group. Crabot et al. (2020) suggest that invertebrate species in historically intermittent streams that have undergone long periods of drying often exhibit specific resistance and resilience traits that enable the community to recover rapidly after rewetting. Contrary to the invertebrate communities, newly intermittent streams may not exhibit these adaptations, potentially leading to more dramatic effects of drying. For example, the Life-history trait group showed significant differences in 4 out of 5 diversity metrics between the 2 water types, highlighting the importance of Life-history strategies in shaping community composition in lowland streams. Furthermore, it is important to consider how these factors contribute to overall stream health and ecosystem function. Although some evidence to the contrary exists (e.g., Carey et al. 2021), some indicates that Life-history traits may respond to flow intermittency (Bêche et al. 2006, Schriever et al. 2015).

Response at trait state level

The differences in trait groups between intermittent and perennial streams are mainly driven by changes in the CWM values of associated traits and trait states, and hence the trait states that primarily determine the functional composition become detectable.

The trait states responsible for differences in trait groups predominantly consist of resilient trait states, particularly evident in response (Ecological) traits like null (cuv1) and slow (cuv2) current velocity preference. Supporting previous observations, taxa that prefer lower velocities were more frequent along the intermittency gradient and in more lentic hydrological conditions (Horrigan and Baird 2008).

For stable (Biological, Dispersal, and Life-history) traits, the proportion of resilient trait states also increases primarily, but because of the inherent complexity and interdependence of traits, the proportion of some related states also increases. Several trait states predominantly in dominant snails (e.g., *Anisus* sp., *Planorbis* sp.) serve as an example of this phenomenon in lowland streams. These snails exhibit trait states such as diapause and dormancy alongside tegument respiration, which are resistant to flow intermittency. Consequently, because of the interconnections among traits, the prevalence of all trait states of these dominant snails increased within the intermittent group. The prominent role of resilient traits in intermittent streams is a well-documented phenomenon in highlands (Falasco et al. 2021), and our results confirm this also in lowland streams. In perennial streams, we found no response (Ecological) traits with a higher proportion, which can be explained by stable hydrological conditions that did not result in significant shifts in CWM values. Notably, stable trait states, including reproduction with clutches or free (rep5), swimmer (lsr3), dispersal ability (fwl1, fwl2), larva aquatic stages (aq52), and those lacking a persistent formula (ref5), confer greater advantages to individuals inhabiting environments characterized by constant hydrological conditions (Gallardo et al. 2014).

Overall, the 3-level analysis revealed a hidden relationship: differences occurring at the state level do not always manifest at the trait group or whole community levels. The true causes, extent, and pathways of these differences still need to be uncovered. One plausible explanation might be that organisms respond to intermittency at the individual level, but individuals possess states from all 4 trait groups, which can be considered trait complexes. Because of the interconnectedness of the trait states, the proportion of states not only might be sensitive to intermittency decrease, but other state

proportions may also change. Another cause might be functional redundancy, which can vary even at the trait group level. Intermittency seems to decrease trait redundancy, but the extent and effect of decreasing function depend on the natural level of trait redundancy in perennial streams.

Each trait group (Ecological, Biological, Dispersal, and Life-history) plays a crucial role in shaping the structure, dynamics, and functionality of macroinvertebrate communities in aquatic ecosystems. Ecological and Dispersal traits allow individuals to respond to changing environmental conditions and enhance community resilience. Biological traits help individuals employ resistance strategies to survive harsh intermittent conditions. Life-history traits support species persistence during disturbances and facilitate rapid recovery and recolonization, maintaining ecosystem functionality and overall enhancing the resilience and stability of the entire community. They influence how species interact with their environment, utilize resources, and respond to changes, ultimately affecting the overall health and stability of the ecosystem.

Conclusion

Our study highlights the need for comprehensive investigation of different aspects related to functional response to flow intermittency. Although our results revealed substantial variations between individual trait groups, we found no overall significance at the whole functional community level. The phenomena that change at the state or trait group level are not always manifested at higher levels, highlighting the importance of considering complex interactions and dynamics within functional communities. The Life-History trait group showed the greatest variation while community differences were most pronounced in the Ecological trait group.

Evaluating an aquatic ecosystem at lower functional levels (trait group and trait state) allows the detection of functional changes that may not be apparent at the whole community level. This 3-level perspective enables researchers to identify changes at the trait group level in the initial stages before they manifest at the whole community level. Exploring the potential of using trait groups as indices and incorporating them into assessment frameworks could facilitate the early detection of functional changes during bioassessment. This approach provides policymakers with an opportunity to intervene early in the process, resulting in more effective conservation and management strategies. Further research is needed to understand the responses of these trait groups and their impact on ecosystem functioning in intermittent streams

across different biogeographical locations and water types. Long-term studies capturing seasonal and interannual variations in community composition and functional traits are also crucial for understanding the dynamics between intermittent and perennial streams. Our findings are particularly relevant in the context of climate change, during which altered precipitation patterns and increased droughts are expected to impact stream ecosystems.

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Data availability

https://github.com/varbirog/functional_diversity_in_macroinvertebrates.

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